ST MARY'S DSG, KLOOF

GRADE: 10

17 October 2025

PHYSICAL SCIENCE CYCLE TEST
KINEMATICS AND ELECTRICITY

TIME: 55 MINS (incl. reading time)
ASSESSOR: T. Kirsten

TOTAL: 63
MODERATORS: K. Bolton and G. Main

NAME: MEMO

SCIENCE TEACHER: _____

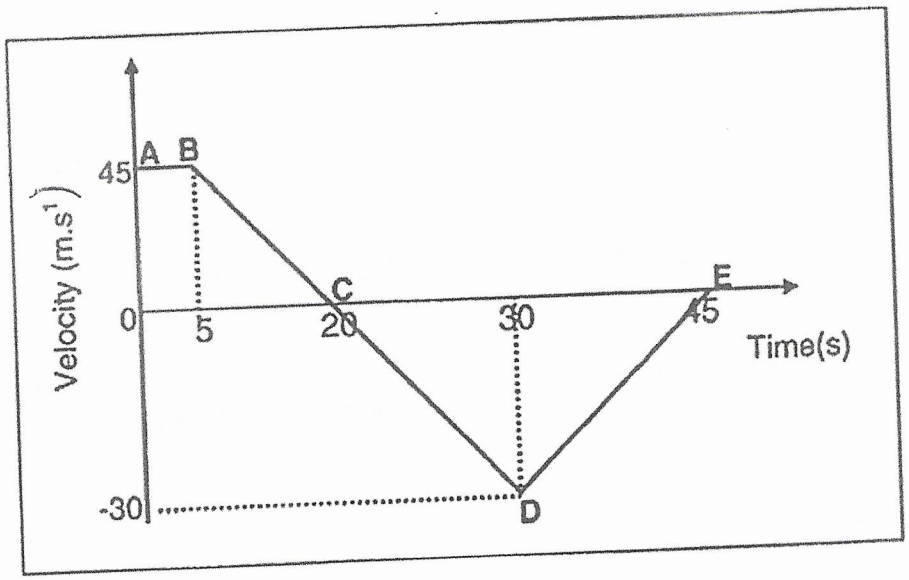
PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 8 pages. A data sheet is provided.
2. Answer on the question paper in the spaces.
3. Write your name and your science teacher's name on your question paper.

SECTION A: KINEMATICS

QUESTION 1

The velocity-time graph of a train initially moving **in a SOUTHERLY direction** is shown below.



1.1 Describe the motion of the train for the following time intervals, in terms of velocity and acceleration. (8)

1.1.1 B to C

Decreasing velocity ✓ South ✓ uniform acceleration North ✓

1.1.2 C to D

Increasing velocity ✓ North ✓ uniform acceleration North ✓

1.2 Without the use of equations of motion, calculate the total displacement for the time interval A-E. (6)

$\Delta x = [(5 \times 45) + (\frac{1}{2} 15 \times 45)] - [(\frac{1}{2} 10 \times 30) + (\frac{1}{2} 15 \times 30)]$

$= 562,5 - 375$

$= 187,5\text{m}$ ✓ South ✓

4.4 Calculate:

4.4.1 the reading on the ammeter (2)

$I = \frac{V}{R} = \frac{18}{9}$ ✓ $= 2A$ ✓

4.4.2 the time it takes for 17 C of charge to pass a point in the above circuit. (3)

$t = \frac{Q}{I}$ ✓ $= \frac{17}{2}$ ✓ $= 8,5s$ ✓

4.5 Define *potential difference*. (2)

The work done per unit positive charge ✓

4.6 Calculate:

4.6.1 the reading on the voltmeter (3)

$V = IR$ ✓ $= 2 \times 3$ ✓

$= 6V$ ✓

4.6.2 the work done in moving 5 Coulombs of charge through the 6 Ω resistor. (4)

$V_{6\Omega} = 18 - 6 = 12V$ ✓

$W = V \times Q$ ✓

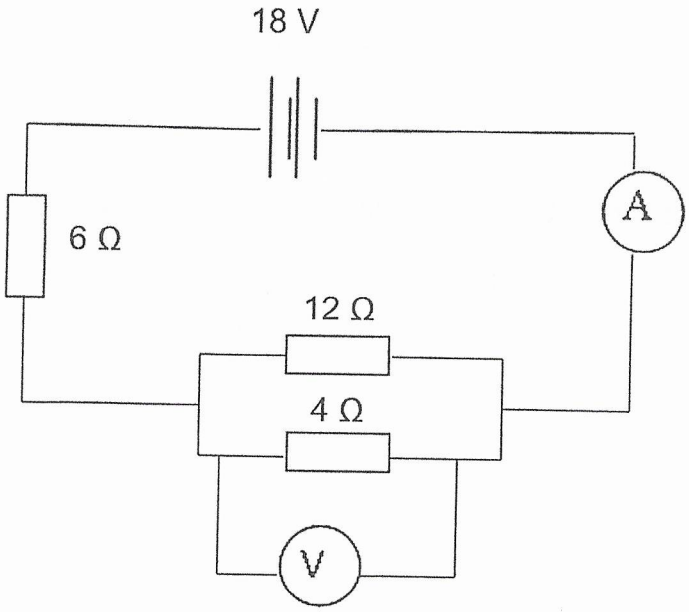
$= 12 \times 5$ ✓

$= 60J$ ✓

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QUESTION 4

A 18 V battery is connected in a circuit as shown in the diagram below.



4.1. Define *resistance*. (2)

A material's opposition to the flow of electric current ✓✓

4.2 Calculate:

4.2.1 the resistance of the parallel combination (3)

$R_{//} = \frac{\text{product}}{\text{sum}} = \frac{12 \times 4}{12 + 4}$ ✓ ✓

$= \frac{48}{16} = 3\,\Omega$ ✓

4.2.2 the total resistance of the given circuit (2)

$3\,\Omega + 6\,\Omega$ ✓ $= 9\,\Omega$ ✓

4.3 Define *current*. (2)

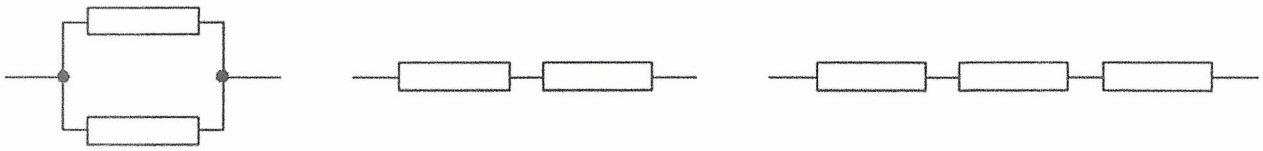
The rate of flow of charge ✓✓

SECTION B: ELECTRICITY

QUESTION 3 MULTIPLE CHOICE Circle the most correct letter

3.1 Identical resistors are connected together to form arrangements X, Y and Z.

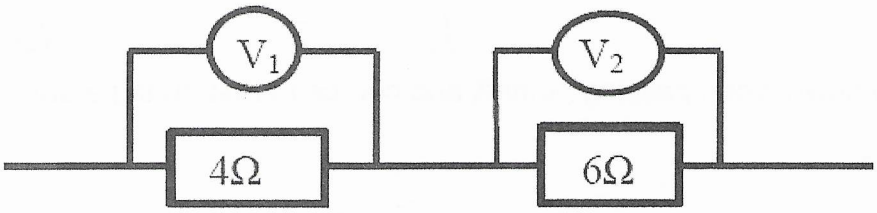
What is the correct order of the resistances of the arrangements from the **largest to the smallest?**



X Y Z

A $X \rightarrow Y \rightarrow Z$
B $Y \rightarrow X \rightarrow Z$
C $Z \rightarrow X \rightarrow Y$
D $Z \rightarrow Y \rightarrow X$

3.2 Consider the diagram below:



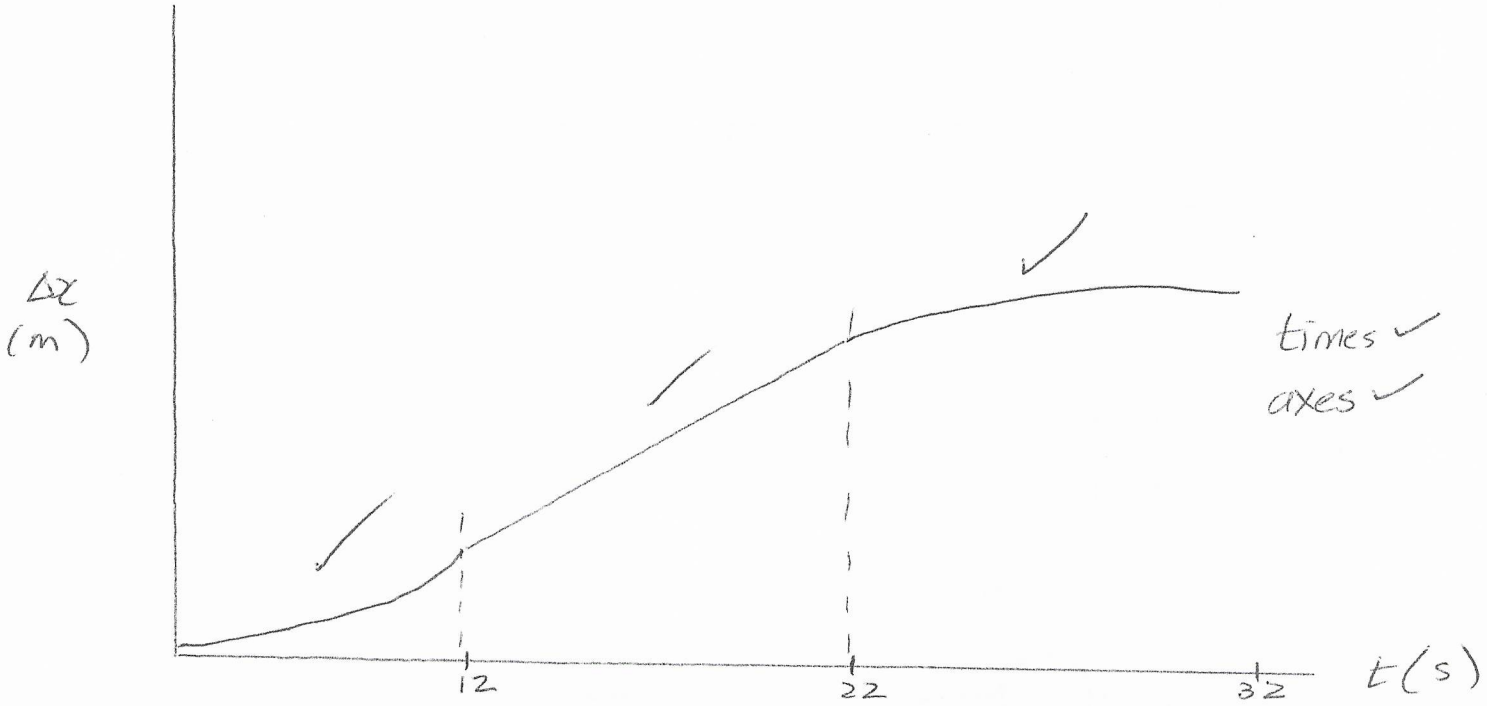
If a current is flowing through both resistors;

A V_1 will have a higher reading than V_2
B V_2 will have a higher reading than V_1
C the reading on V_1 and V_2 will be the same
D potential difference is inversely proportional to resistance.

2 x 2 = [4]

- Athlete **A** accelerates at 0,4 $m \cdot s^{-2}$ for 12 s.
- She then continues to run to the left at constant velocity for a further 10 s.
- She then comes to rest over another 10 s.

2.3 In the space below, draw a position-time sketch graph for athlete **A** for the full 32 s of her run. Take left as positive and use the start as the reference point.
No values need to be shown on the y-axis **but you should label times on the x- axis.** (5)



Athlete **B** decides to slow down when passing point **X** and comes to rest during a time of 7 s.

2.4 Define *acceleration*. (2)

The rate of change of velocity ✓

2.5 Calculate the acceleration of athlete **B** during this time period. (4)

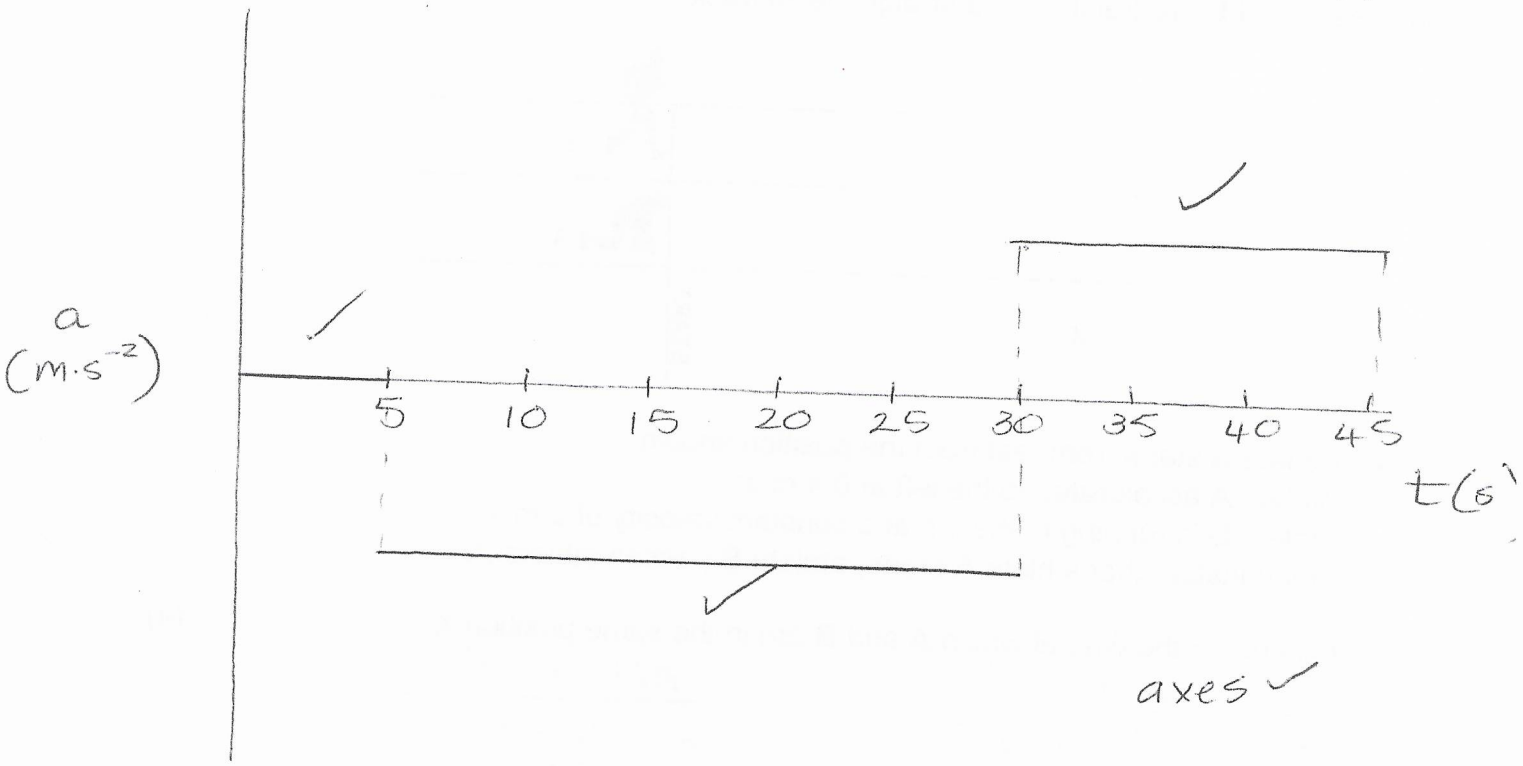
$v_i = 2$
$v_f = 0$
$t = 7s$
$a = ?$

$v_f = v_i + a\Delta t$
$0 = 2 + a\,7$ ✓
$\therefore a = \frac{-2}{7}$
$= -0,29\ m \cdot s^{-2}$
ie $0,29\ m \cdot s^{-2}$ ✓ right ✓

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1.3 Sketch an acceleration vs time graph for the train over the entire journey. Indicate all relevant time values.

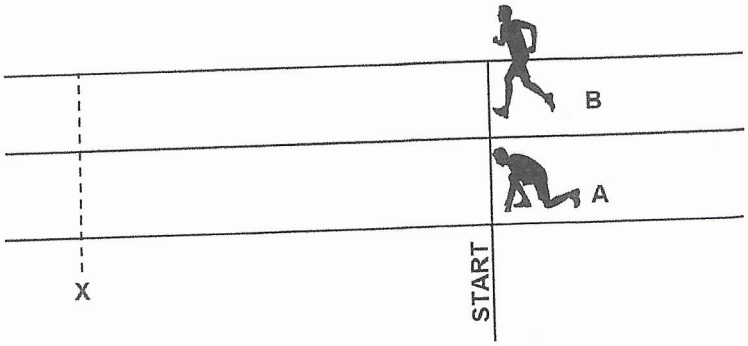
(4)



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QUESTION 2

Athletes **A** and **B** are training on a straight, level track.



- Athlete **A** starts from rest from the position shown.
- Athlete **A** accelerates to the left at 0,4 $m \cdot s^{-2}$.
- Athlete **B** is running to the left at a constant velocity of 2 $m \cdot s^{-1}$.
- At the instant that athlete **A** starts, athlete **B** passes athlete **A.**

2.1 Determine the time at which **A** and **B** are in the same position **X**. (4)

Athlete A

$X = v_i \Delta t + \frac{1}{2} a \Delta t^2$ ✓

$= 0 + 0,2t^2$ ✓

$= 0,2t^2$

Athlete B

$X = v \times t$

$= 2t$ ✓

$0,2t^2 = 2t$ ✓

$\therefore 0,2t^2 - 2t = 0$

$\therefore \quad t = 10\,s$ ✓

2.2 Determine the magnitude of the displacement of the athlete **A** at point **X**, taking the start as the reference position. (3)

$X = v_i \Delta t + \frac{1}{2} a \Delta t^2$

$= 0 + \frac{1}{2}(0,4) \times 10^2$ ✓ ✓

$= 20m$ ✓